EMNLP 2026

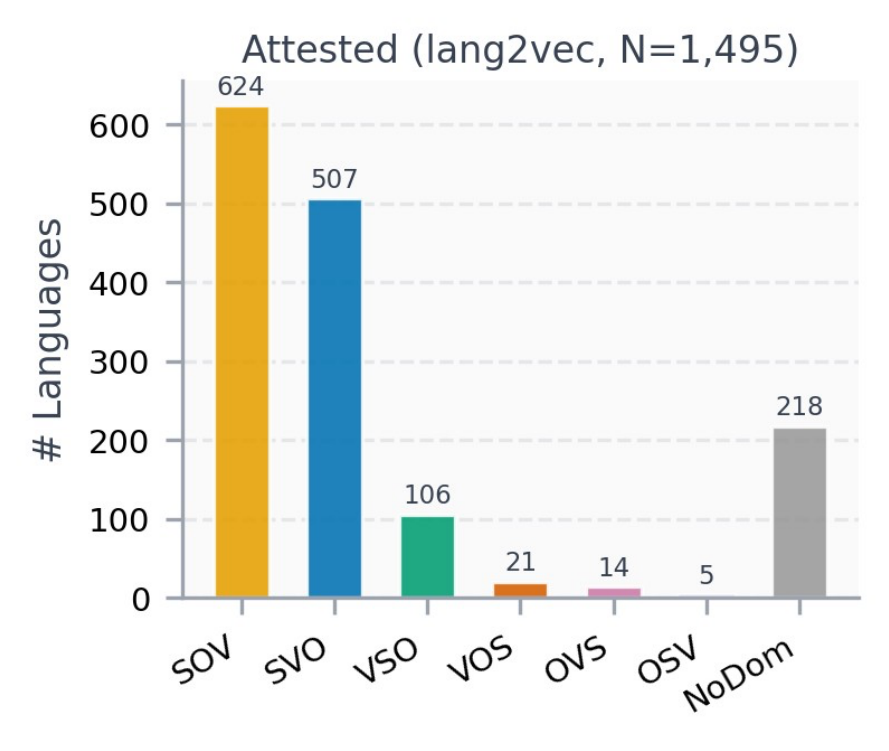
Left-Branching Transformers Excel at Right-Branching Languages

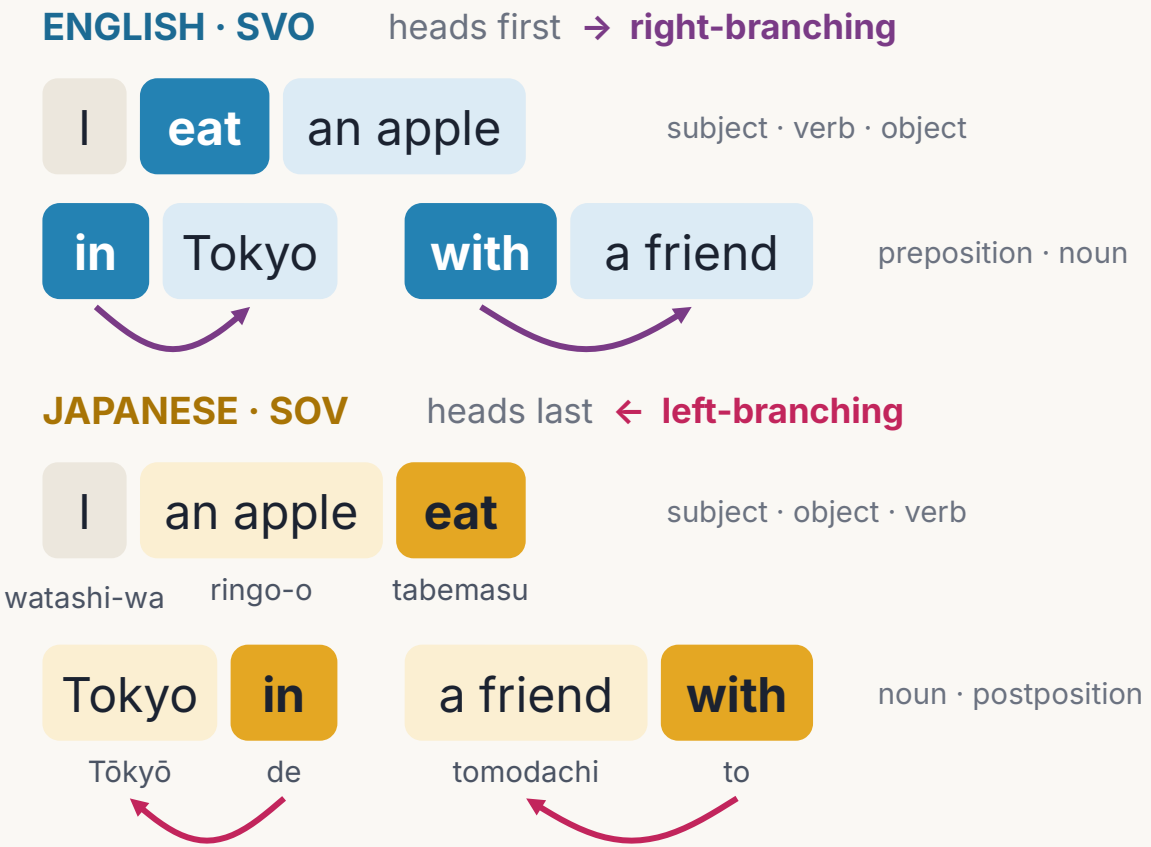
Data Shapes Word Order Preferences in Language Models

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TU Wien · Northeastern University

Natural Languages Vary a Lot in their Word Order





Do LMs Handle All Word Orders Equally Well?

RESEARCH QUESTIONS

RQ1 Do LMs have systematic word order preferences?

RQ2 If there are systematic preferences, are these architectural or data-driven?

RQ3 Do LM preferences mirror attested patterns?

RQ4 Do LM preferences on artificial languages match those on natural languages?

Prior work: mixed results, partly because each study looks at only **one setting**

White & Cotterell 2021 · Kallini et al. 2024 · Kuribayashi et al. 2024 · El-Naggar et al. 2025 · Ziv et al. 2026

Artificial languages only

Controlled, but no semantics, morphology or discourse: findings may not transfer

Natural languages only

Realistic, but word order is confounded with everything else

→ We do both: same architecture, matched training data scales

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Experimental Setup

	Natural languages	Artificial languages
Languages	>70 typologically diverse languages Same language set at every data size	192 PCFG-generated languages 6 base orders × 2^5 internal constituent orderings
Models	Monolingual **Goldfish** (GPT-2 architecture) + multilingual mGPT, XGLM, BLOOM	**GPT-2 small**, trained from scratch Matched to Goldfish at 5 & 10 MB
Training data	Web text: word order + everything else 5 MB → 1 GB per language, byte-premium scaled	Identical data, only word order differs 5 & 10 MB · shared word-boundary BPE tokeniser
Evaluation	FLORES-200 & PUD BPEC (bits per English character)	Held-out sentences Perplexity

$$\mathrm{BPEC}_\ell = \frac{-\sum_{s \in S_\ell} \sum_{t=1}^{n_s} \log_2 p_\theta\left(x_t^{(s)} \mid x_{<t}^{(s)}\right)}{C_{\mathrm{eng}}}$$

BPEC: bits per English character

NLL ÷ English character count: comparable across tokenisers

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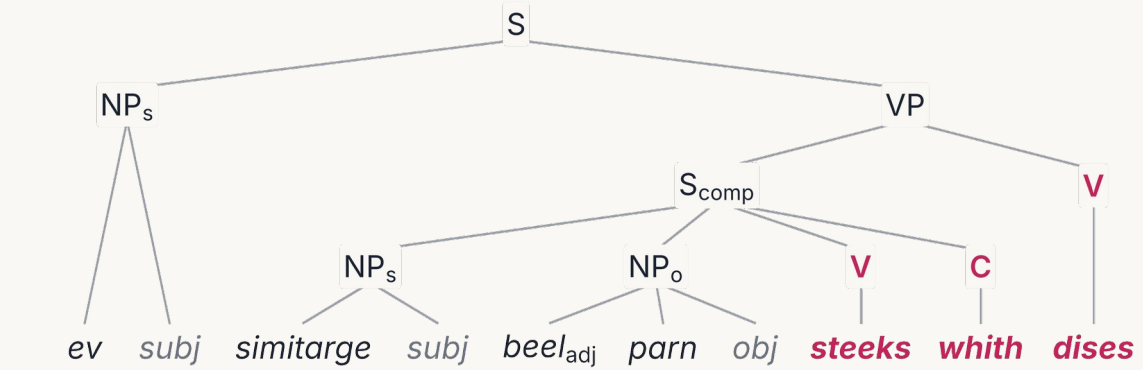
Shared across both settings: **GPT-2 architecture · 5 & 10 MB of training data**

Building Artificial languages

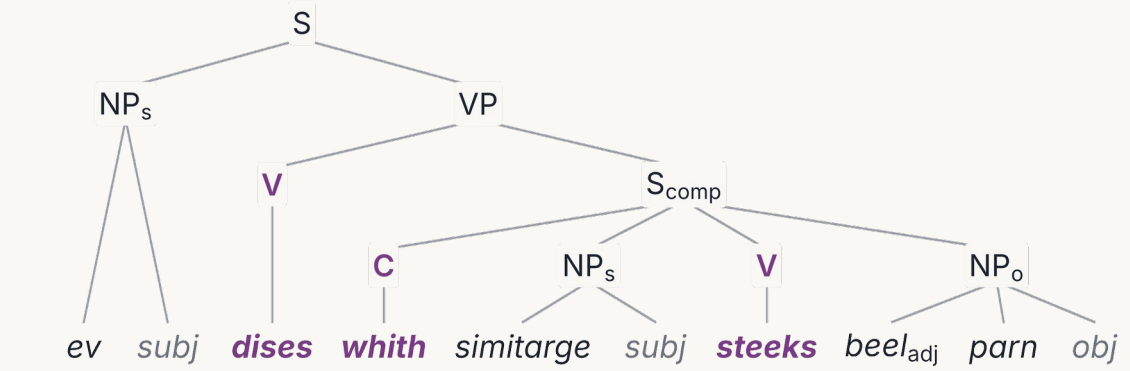
Switch	L	R
1 **Complement clause**	S-comp V	V S-comp
2 **Complementiser**	S Comp	Comp S
3 **Adposition**	NP Postp	Prep NP
4 **Adjective**	Adj N	N Adj
5 **Relative clause**	VP Rel N	N Rel VP

6 base orders × 2^5 = 192 languages

SOV-LLLLL ~ Japanese



SVO-RRRLR ~ English



Labeling Natural Languages the Same Way

ONE LABEL PER LANGUAGE

SVO- **R** **R** **R** **L** **R**

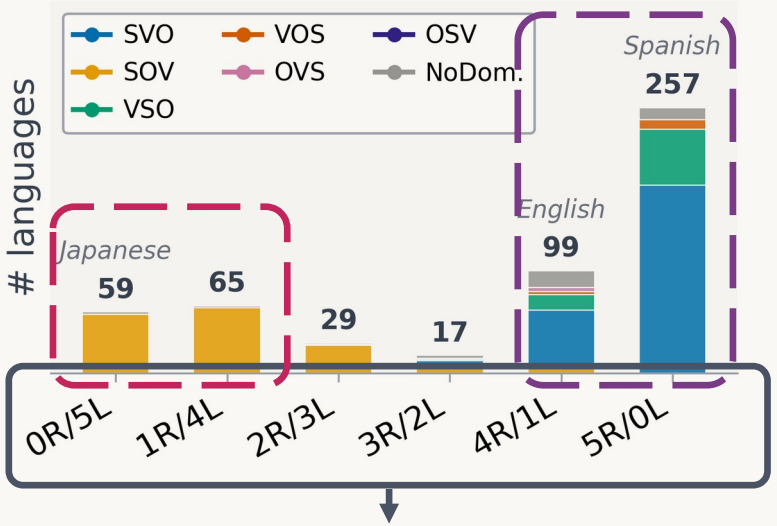
1 complement clause | **2** comple-mentiser | **3** adposition | **4** adjective | **5** relative clause

base word order + 5 binary switches: **L** left- / **R** right-branching

Language	Label	Scale
English	SVO-**RRRLR**	4R/1L
French	SVO-**RRRRR**	5R/0L
Hindi	SOV-**LRLLL**	1R/4L
Japanese	SOV-**LLLLL**	0R/5L

Labels from WALS, Grambank and APiCS; only languages with all five switches annotated

Documented natural language distribution (N=526)



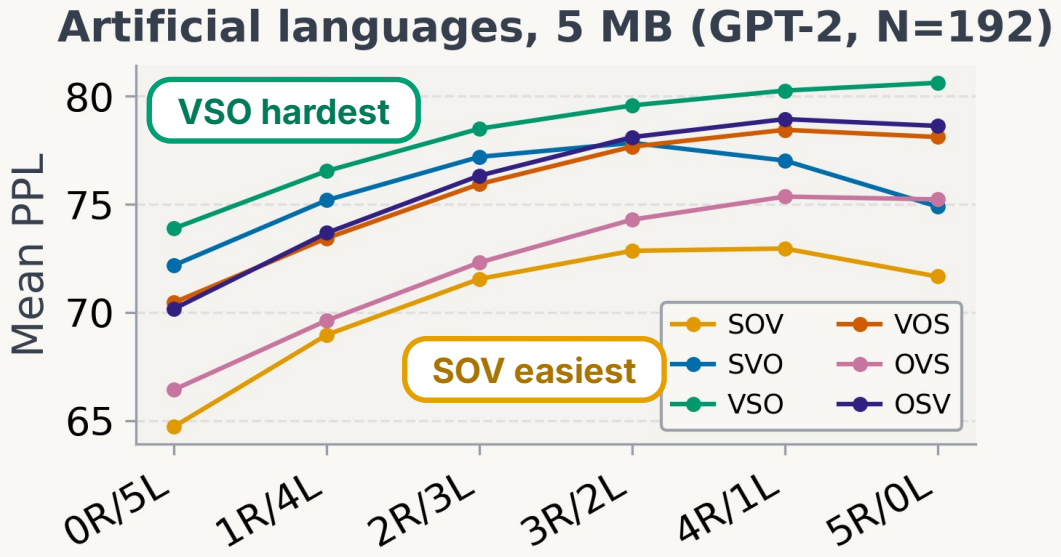
Scale = number of right-branching switches
0R/5L fully left-branching · **5R/0L** fully right-branching

Natural languages cluster at both extremes:
mostly left- and **mostly right-branching**

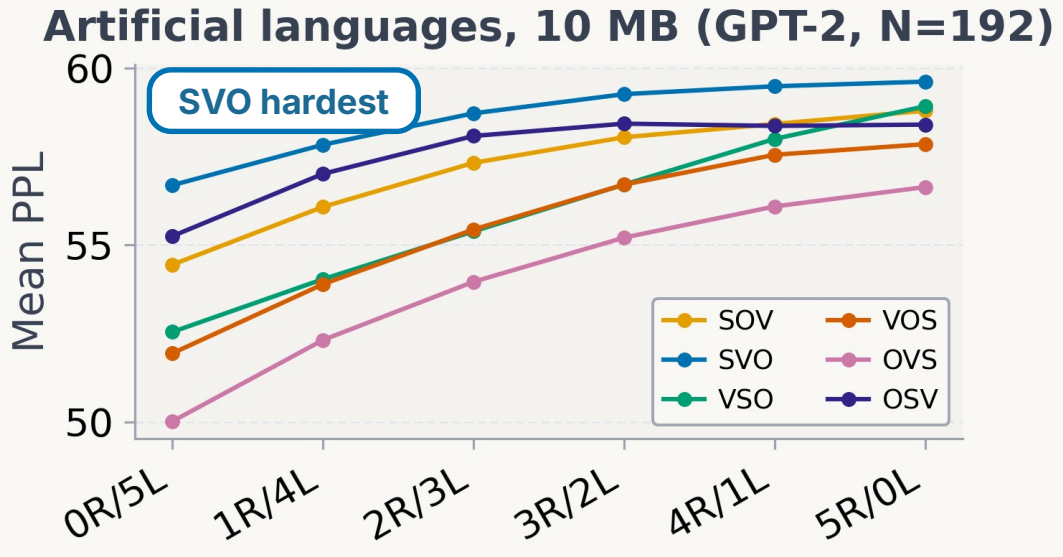
LM Preferences in Artificial Languages

Base word order rankings flip with more training data

GPT-2 · 5 MB



GPT-2 · 10 MB



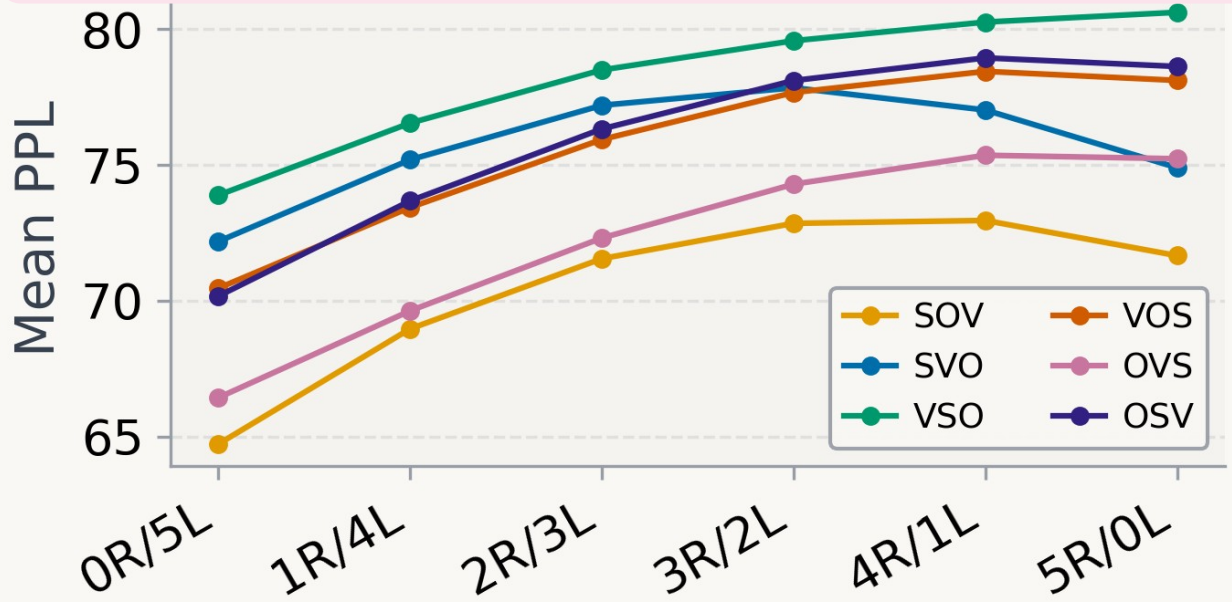
5 MB: **SOV** easiest, **VSO** hardest → **10 MB:** VS orders easiest, **SVO** **hardest**

No stable base word order preference; at scale on natural languages, **SVO** will be the **easiest**

LM Preferences in Artificial Languages

Left-branching is easiest for all six base word orders, consistently at 5 and 10 MB

Left-branching preference also reported in prior work
White & Cotterell, 2021; Kuribayashi et al., 2024; El-Naggar et al., 2025

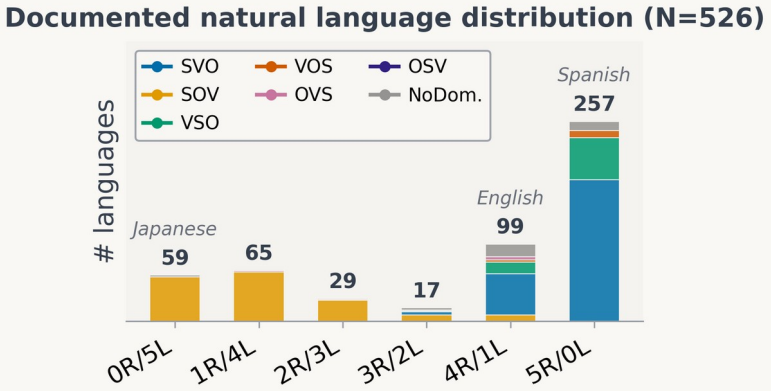


← fully left-branching
e.g. SOV-LLLLL ≈ Japanese

fully right-branching →
e.g. SVO-RRRRR ≈ French

Mean PPL over 10 runs · lower = easier

Natural languages: SOV mostly left-, SVO mostly right-branching



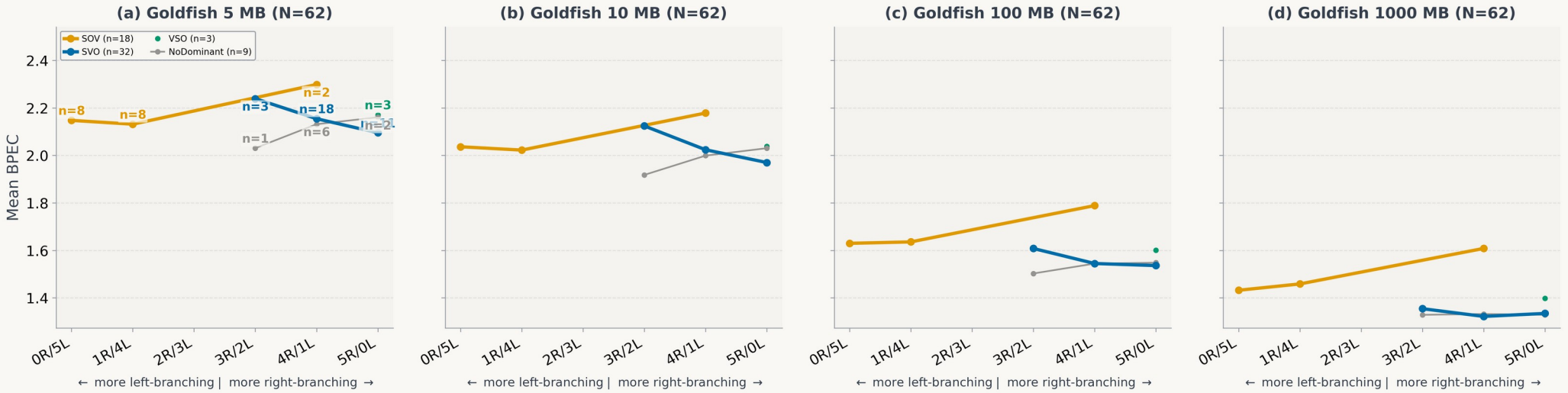
LMs show no such pattern on artificial languages: left-branching is easiest for every base word order

The easiest configuration for each base word order has **no natural counterpart**

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LM Preferences in Natural Languages

No preference at 5 MB. With more data, right-branching SVO becomes **easiest**.



SVO–SOV δ	0.00	−0.10	**−0.60*** **	**−0.76*** **

Opposite of artificial languages: there, SVO is the **hardest** base word order at 10 MB; here, the **easiest** at scale

Same languages at every size: by 1 GB, SOV is ≈10% harder to model, though it is the most frequent order worldwide

Also holds for multilingual models (XGLM, mGPT) · details in the paper

Mean BPEC on FLORES-200, lower = easier · δ = Cliff's δ, < 0: SVO easier · *** p < .001

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What Drives the SVO Advantage?

Linear mixed-effects model: the growing SVO–SOV gap survives every control

> **Not script.** 71% of SOV vs 19% of SVO FLORES-200 languages are non-alphabetic, but on alphabetic scripts only the gap still grows with data

> **Not morphological complexity.** Its effect shrinks with data, so it cannot explain a gap that grows

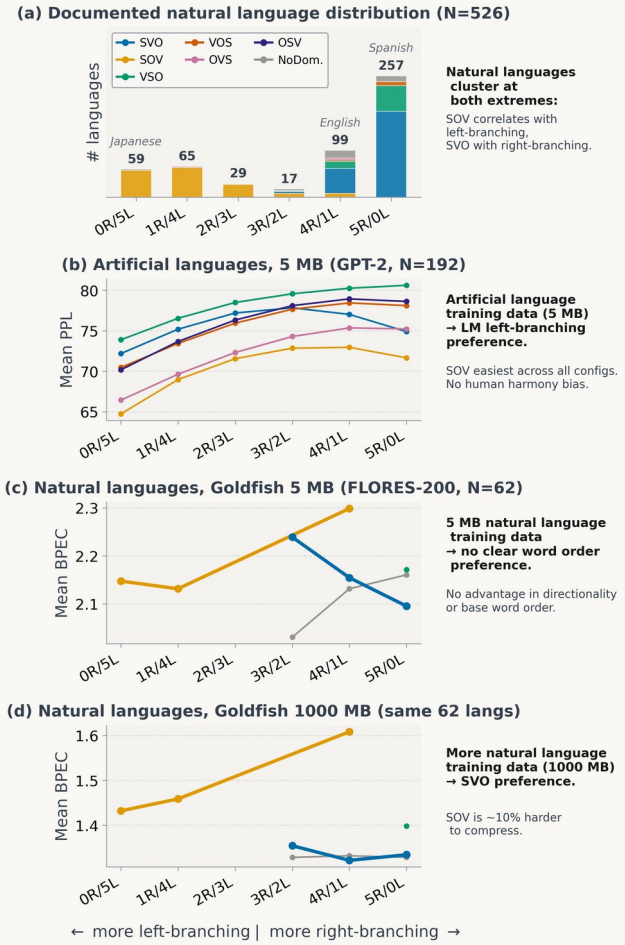
> **Not training data composition.** Web-crawl (OSCAR) vs. curated share of each language's data: no effect on the growth

> **Partly language resourcedness** (≈20% of the effect): a proxy for data quality

Linear mixed-effects model, 50 SVO/SOV languages × 4 sizes: $\text{BPEC} \sim \text{word order} \times \text{size} + \text{covariates} + (1 \mid \text{language})$

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Takeaways



1 **Systematic, but not human-like: left-branching** on **artificial languages** regardless of base word order and against word order correlations in **natural languages**

2 **Data overrides the architectural bias:** on **natural languages** at scale, better-resourced, **right-branching SVO** languages become easiest, in mono- and multilingual models

3 **LM preferences on artificial and natural languages go in opposite directions at scale: SVO** is hardest on **artificial languages**, easiest on **natural ones**

Even monolingual models trained on just 1 GB develop the **SVO** preference; frontier models, with far larger resource gaps between languages, likely amplify it.

A precondition for ***silent typological homogenisation***: flexible word order languages drifting towards the dominant ones.

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